Water Resources – Sources and Sustainability

MODULE 3 LECTURE 2

Sustainable water use

- Sustainable water use refers to the management of water resources in a way that meets the needs of the present without compromising the ability of future generations to meet their own needs.
- Principles of Sustainable Water Use
  - Efficiency: Maximizing water use with minimal waste.
  - Equity: Ensuring that all sectors (domestic, agriculture, industry) have fair access to water resources.
  - Environmental Protection: Maintaining healthy ecosystems that depend on water, such as wetlands, rivers, and aquifers.
  - Resilience: Ensuring that water resources can withstand and recover from stresses such as droughts, floods, and pollution.

Importance of Sustainable Water Use

- Resource Scarcity: With growing water demand, especially in arid and semi-arid regions, sustainable management ensures continued access to freshwater.
- Climate Change: Unpredictable weather patterns increase the need for adaptive water management.
- Public Health: Clean water is essential for health, and sustainable practices help maintain water quality.

Key Strategies for Sustainable Water Use

- Water Efficiency Improvements: Implementing technologies and practices that reduce water consumption.
- Wastewater Treatment and Reuse: Recycling and reusing water for non-potable uses (irrigation, industrial cooling).
- Integrated Water Resource Management (IWRM): A holistic approach that considers all water users (urban, rural, agricultural) and integrates environmental, economic, and social aspects.
- Rainwater Harvesting: Collecting and storing rainwater for later use, especially in water-scarce areas.
- Pollution Control: Reducing pollution through improved waste management and treatment facilities.

Strategies for water conservation and sustainable water use in various sectors

Watershed Management

- To control damaging runoff and degradation and thereby conservation of soil and water.
- To manage and utilize the runoff water for useful purpose.
- To protect, conserve and improve the land of watershed for more efficient and sustained production.
- To protect and enhance the water resource originating in the watershed.
- To check soil erosion and to reduce the effect of sediment yield on the watershed.
- To rehabilitate the deteriorating lands.
- To moderate the floods peaks at down stream areas.
- To increase infiltration of rainwater.
- To improve and increase the production of timbers, fodder and wildlife

Watershed Management Practices

- Vegetative measures (Agronomical measures)
  - Strip cropping
  - Pasture cropping
  - Grass land farming
  - wood lands
  - Establishment of permanent grass and vegetation
  - Providing vegetative and stone barriers
- Engineering measures (Structural practices)
  - Contour bunding
  - Terracing
  - Construction of earthern embankment
  - Construction of check dams
  - Construction of farm ponds
  - Construction of diversion
  - Gully controlling structure
  - Rock dam
  - Construction of silt tanks

Agriculture

- Agriculture accounts for about 70% of global freshwater use, mainly for irrigation

- Drip Irrigation: Delivers water directly to the plant roots, reducing evaporation and runoff.
- Drought-Resistant Crops: Using crops that require less water.
- Efficient Water Management: Proper timing and quantity of irrigation based on weather conditions and soil moisture

Industry

- Industry accounts for 20% of water use, often in manufacturing, cooling, and cleaning processes

- Closed-Loop Systems: Reusing water within the system to minimize consumption.
- Water Recycling: Treating and reusing wastewater for industrial processes or cooling.
- ZLD – Zero liquid discharge

Urban Areas – Domestic use

- Accounts for 8% of water use
- Leak Detection and repair in water supply systems.
- Low-flow fixtures (e.g., low-flow faucets, showerheads, and toilets).
- Rainwater Harvesting: Collection of rainwater for domestic use, reducing reliance on external sources.
- Greywater Recycling: Reusing water from sinks, showers, and washing machines for non-potable purposes like irrigation.
- Smart Metering and Monitoring: Use of IoT and data analytics to detect leaks, monitor usage patterns, and optimize water distribution.

Water Conflicts

Water Conflicts

- Competition for Limited Resources: When water is scarce or unevenly distributed, competition between sectors (agriculture, industry, domestic) increases.
- Transboundary Water Issues: Shared rivers and aquifers often lead to conflicts between countries or regions over water rights and usage.
- Pollution: Contamination of water sources can lead to disputes over water quality and access.
- Climate Change: Changes in rainfall patterns and rising temperatures exacerbate water scarcity, leading to conflicts over access to available water resources.

Impact of Water Conflicts

- Social and Political Instability: Conflicts over water can lead to regional tensions and even military conflicts.
- Economic Consequences: Water scarcity can lead to reduced agricultural yields, industrial output, and power generation, impacting economies.
- Environmental Degradation: Over-extraction of water can harm ecosystems, causing river drying, habitat loss, and biodiversity reduction

Global Case Studies

Nile River Conflicts

ETHIOPIA – BACKGROUND

- Treaties regarding Nile River water rights date back to the colonial period and continue to be referenced today in legal matters.
- In the colonial period of the early 20th century, historically significant treaties were signed between colonial powers and Nile River riparian states, including the Treaty Between Great Britain and Ethiopia, the Tripartite Treaty, and the Agreement between Egypt and Anglo-Egyptian Sudan.
- The treaties significantly undermined Ethiopia's rights to the Nile River, despite the Blue Nile running through Ethiopia's sovereign land, and favored Egypt and Sudan's rights to the Nile's water.

EGYPT AND SUDAN – BACKGROUND

- Both Egypt and Sudan are Lower Nile Basin countries and receive relatively low rainfall levels compared to Upper Nile Basin states, making them highly dependent on the Nile River for water supply.
- According to the UN, Egypt has an annual water deficit of around seven billion cubic meters and is projected to completely water-scarce by 2025.
- Egypt's water scarcity resulted in protests in 2007, known as the "Revolution of the Thirsty," after residents of the Nile Delta reported putrid water in their limited water supply.
- As such, both countries continue to reference these colonial-era treaties in a bid to remain water-secure despite environmental pressures. However, Ethiopia refuses to recognize the

Nile River Conflicts

- Tensions in 1978 following Ethiopia's proposal of dam construction on the Blue Nile.
- Ethiopia's proposal was met with significant Egyptian backlash that ultimately led to the failure of the project.
- In 2010, Ethiopia announced its intention to build the largest and most expensive hydroelectric dam in Africa. The proposal was immediately met with Egyptian and Sudanese backlash, with Egypt calling for African Union (AU) and United Nations (UN) intervention on the matter and then Egyptian Minister of Water and Environmental Resources, Mohamed Nasr Eldin Allam, forming an agreement with the Sudanese government to preserve "the historical rights of both countries to the Nile's water."
- Although Sudan declared neutrality in the dispute in 2014,

Indus River Conflict

GEOGRAPHY

- Glaciers and tributaries originating in the high mountains of the Ngari Prefecture in western Tibet, the Himalayas, Hindu Kush in Afghanistan and the Karakoram feed the extensive Indus river system.
- Its floodplain, where most of Pakistan’s population live, is one of the largest agricultural regions in Asia.
- Around 90% of Pakistan's food and 65% of its employment depend on farming and animal husbandry, which are sustained by the Indus

Indus River Conflict

PAKISTAN CONTEXT

- According to NASA, the Indus Basin is the world's second most over-stressed aquifer
- Over-extraction of its finite groundwater resources is a major challenge in the Indus Basin
- Unlike India, Pakistan relies almost exclusively on the Indus, and southern downstream areas are especially vulnerable to strains on the basin's water supply.

INDIAN CONTEXT

- The position of the LOC meant that India gained control of upstream barrages, which regulated water flow into Pakistan.
- As the boundary between India and Pakistan cut across many of the river's tributaries, an upstream-downstream power structure emerged, which has been the source of tensions between the two countries, particularly in response to dam projects in Indian-administered territory
- Over the past two decades Pakistan has launched multiple attempts to prevent

Indus River Conflict – CONFLICT RESOLUTION

- The Indus Water Treaty (IWT)- signed in 1960, and mediated by the World Bank
- Under the IWT, control over the three eastern tributaries of the Indus River—Ravi, Sutlej, and Beas—is granted to India before they flow into Pakistan, and the three western tributaries—Indus, Jhelum, and Chenab—to Pakistan
- Under the IWT, the Permanent Indus Commission was established, composed of equal numbers of representatives from both India and Pakistan.
- All developments along the Indus must be reported to the other party. Any differences in opinion with regards to interpreting the application of the treaty is first referred to the Commission.
- Should a disagreement emerge, an independent third party may be approached. If the difference in opinion is regarded as a dispute by the neutral party, a court of arbitration can be established to resolve the issue
- IWT has generally been considered a success, surviving multiple interstate

The Colorado River

- United States and Mexico over water rights to the Colorado River.
- The Colorado River provides water to more than 40 million people across seven southwestern states, 29 tribal nations and Mexico
- In Las Vegas, 90% of its water supply comes from the river. In Phoenix and was it's 50% and in Los Angeles it's 25%.
- According to a 1922 agreement, each of these seven states have a legal right to a certain amount of the river's water. But this compact was made under the assumption that there was more water than there actually was.
- Scientists have been predicting for years that the Colorado River would continue to deplete due to global warming and increased water demands
- Between January 2000 and April 2023, the amount of water stored in Lake Mead and Lake Powell, the two largest reservoirs in the United States, declined by 33.5 million acre feet (41.3 billion cubic meters).

Interstate River disputes in India

- Krishna water dispute between states of Maharashtra, Karnataka and Andhra Pradesh (old)
- Godavari water dispute between states of Maharashtra, Andhra Pradesh, Karnataka, Madhya Pradesh and Goa
- Narmada water dispute between the states of Rajasthan, Madhya Pradesh, Gujarat and Maharashrtra
- Cauvery water dispute between the states of Karnataka, Kerala, Tamil Nadu and Union Territory of Pondicherry
- Mahadayi/Mandovi water dispute between the states of Goa, Karnataka and Maharashtra
- https://blog.ipleaders.in/krishna-water-dispute-an-overview/

Enlist and explain the major water conflicts around the world